TREEWALK NYC • STEAM LESSON PLAN

Japanese Tree Lilac

Syringa reticulata

Species background: Native to Japan, Korea, and parts of northern China; despite its common name, it's a true member of the lilac genus (Syringa), grown as a small tree rather than a shrub. It's the only true tree-sized member of the lilac group commonly planted on city streets -- most other lilacs are large shrubs, not trees, which is what makes this one useful as a small street tree.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Japanese Tree Lilac: Identify this tree as a lilac and find its cream-colored flowers.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

I'm part of the lilac family, but I grow as a real tree instead of a bush! Smell my creamy white flowers in early summer -- they bloom later than other lilacs.

Procedure

1. Engage: Ask if students know what a lilac shrub looks like.
2. Explore: Look at this tree's size and compare it to a shrub-sized lilac if known.
3. Explain: Explain that this is a lilac that grows as a tree instead of a bush.
4. Art: Smell the flowers, then draw swirls to show what the smell feels like.
5. Evaluate: Student can describe this tree as a tree-shaped lilac.

Discussion questions

What other plants do you know that come in both bush and tree forms?

Assessment

Correctly describes the tree as a tree-form lilac.

Extension

Find a lilac shrub elsewhere and compare its size to this tree.

GRADE BAND 3-5

Japanese Tree Lilac: Explain how this tree's bloom timing extends the lilac flowering season.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

This tree smells sweet in early summer -- it blooms later than shrub lilacs, which mostly finish in spring.

Procedure

1. Engage: Ask what it would mean for a city to have lilac blooms for a longer stretch of the year.
2. Explore: If in bloom, observe the large flower clusters; otherwise look at photos.
3. Explain: Explain that this tree extends the lilac season into early summer.
4. Art: This tree smells sweet in early summer -- draw what that smell might look like as swirls of color and lines around the flowers.
5. Evaluate: Student explains in one sentence why planting both shrub lilacs and this tree extends lilac season.

Discussion questions

Why might a city want flowering plants blooming across as many months as possible?

Assessment

Correct one-sentence explanation of extended bloom season.

Extension

Research when shrub lilacs typically bloom compared to this tree.

GRADE BAND 6-8

Japanese Tree Lilac: Explain family-level plant relationships using this tree's true lilac identity as a case study.

Standards connection: NGSS MS-LS4-2; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, sketching materials.

Teacher background

This tree really is a true lilac (genus Syringa) -- just grown as a small tree instead of a multi-stemmed shrub, with flat-topped white flower clusters instead of purple ones. It's a cousin of ash trees in the olive family (Oleaceae).

Procedure

1. Engage: Ask what surprises students about this tree being 'really' a lilac.
2. Explore: Compare this tree's bark and leaf shape to a nearby green ash's.
3. Explain: Explain that plant classification is based on shared floral and fruit structure, not size or growth habit, which is why this tree and green ash are family cousins.
4. Art: Create a synesthesia-inspired painting translating the scent of the flowers into color and shape, then write a short explanation of your choices.
5. Evaluate: Write a paragraph explaining why this tree is classified as a true lilac despite looking so different from a lilac shrub.

Discussion questions

Why might plant classification based on flower and fruit structure be more reliable than classification based on overall size or shape?

Assessment

Paragraph correctly explains classification by floral/fruit structure, not appearance.

Extension

Compare this tree's bark and leaf arrangement directly to a nearby green ash.

GRADE BAND 9-12

Japanese Tree Lilac: Research one shared floral trait uniting the olive family and check it against a nearby ash tree.

Standards connection: NGSS HS-LS4-1; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, sketching materials.

Teacher background

Japanese tree lilac, common lilac, ash, forsythia, and privet are all in the olive family (Oleaceae) -- a useful case for why plant classification is based on shared floral and fruit structure rather than overall size or growth habit. Research one shared floral trait (such as opposite leaves or four-petaled tubular flowers) that unites this family and check it against a nearby ash tree.

Procedure

1. Engage: What kind of evidence would convince a botanist that two very different-looking plants belong in the same family?
2. Explore: Research the defining floral traits of the olive family (Oleaceae).
3. Explain: Test one of those traits (such as opposite branching or flower structure) against a nearby green ash tree.
4. Art: Research Wassily Kandinsky's color-sound theories, a real synesthetic approach used in early abstract art, then apply a similar cross-sensory translation method to this tree's scent, documenting your process and color choices.
5. Evaluate: Submit the family-trait research, the direct comparison to ash, and the synesthesia-inspired artwork with process notes.

Discussion questions

How does this case illustrate the difference between classification based on deep shared ancestry versus classification based on surface-level resemblance?

Assessment

Rubric covering accuracy of the family-trait research and depth of the cross-sensory art process.

Extension

Research one other unexpected member of the olive family beyond ash, lilac, forsythia, and privet.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.